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SUPERSEDING
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9 August 1954

MILITARY SPECIFICATION

PLASTIC SHEETS, VINYL, FLEXIBLE, TRANSPARENT, OPTICAL QUALITY

1. SCOPE

1.1 This specification covers the basic properties of a vinyl resin composition, transparent, flexible plastic sheet of substantial thickness intended for optical and other uses.

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL
PPP-D-723 - Drums, Fiber.

MILITARY
MIL-T-19102 - Test Procedures for Mask, Gas, ND Mark V.

NAVY DEPARTMENT
General Specifications for Inspection of Material.

STANDARDS

MILITARY
MIL-STD-105 - Sampling Inspection Table for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

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2.2 Other publications.- The following documents form a part of this specification. Unless otherwise indicated the issue in effect on date of invitation for bids shall apply.

CONSOLIDATED CLASSIFICATION COMMITTEE
Consolidated Freight Classification Rules.

(Application for copies should be addressed to the Consolidated Classification Committee, 202 Chicago Union Station, Chicago 6, Ill.)

AMERICAN SOCIETY FOR TESTING MATERIALS
D412 - Tension Testing of Vulcanized Rubber.
D746 - Method of Test for Brittleness Temperature of Plastics and Elastomers by Impact.
D792 - Methods of Test for Specific Gravity of Plastics.
D1003 - Method of Test for Haze and Luminous Transmittance of Transparent Plastics.

(Application for copies should be addressed to the American Society for Testing Materials, 1916 Race Street, Philadelphia 3, Pa.)

3. REQUIREMENTS

3.1 Material.- The material shall consist of virgin vinyl chloride polymer and copolymer resin suitably plasticized with only phthalate type plasticizers and any necessary stabilizer and lubricants, properly calendered and press planished to produce best optical properties.

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3.2 Property values. - The plastic sheet, when subjected to the tests shown in table I, shall conform to the property values shown therein. Five determinations shall be made for each test (except brittleness temperature). These values shall be averaged to determine conformance with this specification.

Table I - Property values.

Property to be tested	Test paragraph	Unit of value	Value required
Specific gravity	4.5.1	Maximum	1.26 ± 0.02
Tensile strength	4.5.2	Minimum	
lengthwise		p.s.i.	2800
crosswise			2800
Elongation	4.5.3	Minimum	
lengthwise		percent	250
crosswise			250
Brittleness temperature (50 percent brittle point)	4.5.4	Maximum degrees C.	-20
Light transmission A	4.5.5	Minimum percent	78
Light transmission B	4.5.5	Minimum percent	81.0
Haze	4.5.6	Maximum percent	4

3.3 Uniformity. - All sheets of any lot shall be uniform in dimension, finish, and specified properties (see table I).

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3.4 Dimensions and tolerances.-

3.4.1 Length, width and thickness.- Unless otherwise specified (see 6.2), the material shall be the manufacturer's standard size 20 inches by 50 inches, minus 0 inch, plus 1/4 inch. Thickness shall be in accordance with table II.

Table II - Thickness.

Inches	Tolerance
0.040	±10 percent
.060	±10 percent

3.5 Color.- The plastic sheet shall be colorless and transparent.

3.6 Minor optical defects.- Minor blemishes which do not individually reduce visibility through the plastic shall be disregarded, unless they are grouped in an objectionable pattern.

3.7 Workmanship.- The sheets shall be free from blisters, wrinkles, cracks, dents, heat marks, and graininess which will affect the serviceability of the sheet.

4. QUALITY ASSURANCE PROVISIONS

4.1 Inspection.- Inspection shall be made by the Government or under Government supervision to determine compliance with the requirements of this specification.

4.2 Sampling.- An inspection lot shall consist of all sheets of one thickness, of a given machine run and offered for inspection at one time. A machine run shall consist of all the material pressed in which the basic resin and processing conditions are the same. The largest piece of sheet material remaining untested from that selected for test may, at the discretion of the Government inspector, be furnished as part of the contract or order.

4.2.1 Sampling for visual and dimensional inspection.- A random sample of plastic sheets shall be selected from each lot of material offered for Government inspection of visual and dimensional characteristics in accordance with Standard MIL-STD-105 at inspection level III. The AQL shall be 1.5 percent defective.

4.2.1.1 Sampling for special dimension pieces for visual and dimensional inspection shall be in accordance with 4.2.1.

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4.2.2 Sampling for tests.- A random sample of plastic sheets shall be selected from each inspection lot in accordance with table III and shall be subjected to tests specified in table I.

Table III - Sampling for lot acceptance tests.

Number of sheets or special dimension pieces in the lot	Number of sheets or special dimension pieces in the sample
15 and under	1
16 to 65	2
66 to 200	3
201 to 500	4
501 and over	5

4.2.3 At the option of the manufacturer a rejected lot may be resubmitted for lot acceptance inspection and tests, but only after the manufacturer has subjected specimens from each sheet in the lot to the lot acceptance test wherein failure was observed and has removed all nonconforming sheets. The Government inspector shall again select and examine samples from such resubmitted lots to verify compliance with this specification.

4.2.4 Reports of lot acceptance tests.- Acceptance or rejection of lots will normally be determined by the Government inspector on the basis of the lot acceptance tests. Three copies of the lot acceptance test report, certified by the Government inspector, shall be forwarded to the bureau or agency concerned with a statement as to whether the lot was accepted or rejected.

4.3 Inspection.- Each of the sample sheets selected in accordance with 4.2.1 shall be subjected to surface inspection by the Government inspector at the place of manufacture to determine compliance with section 3 except 3.1 and 3.2. Any sheet in the sample containing one or more visual or dimensional defects shall be rejected and if the number of defective sheets in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

4.4 Conditioning of test specimens.- Tests shall be conducted in the "as received" condition; that is, no special conditioning shall be required.

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4.5 Test methods.-

4.5.1 Specific gravity.- Test for specific gravity shall be performed in accordance with Method D792 of the American Society for Testing Materials(ASTM).

4.5.2 Tensile strength.- Test for tensile strength shall be performed in accordance with Method D412 of ASTM, except that tensile strength shall be determined on a pendulum type or constant rate of elongation testing machine of a type approved by the bureau or agency concerned. The machine shall preferably be power-driven. Tensile strength shall be determined in two directions, the lengthwise direction and the crosswise direction.

4.5.3 Elongation.- Tests for tensile elongation shall be performed in accordance with Method D412 of ASTM. Apparatus used shall be as specified in 4.5.2.

4.5.4 Brittleness temperature.- The 50 percent brittleness temperature shall be determined in accordance with Method D746 or ASTM. A motor driven brittleness temperature tester shall be used.

4.5.5 Light transmission.- Tests for light transmission A shall be made in the visible light spectrum range of 400 to 700 millimicrons in accordance with Method D1003 of ASTM. Tests for light transmission B shall be made at 450 millimicrons in accordance with Specification MIL-T-19102.

4.5.6 Haze.- Test for haze in the spectrum range of 400 to 700 millimicrons shall be made in accordance with Method D1003 of ASTM.

4.6 Inspection procedures.- For Naval purchases, the general inspection procedures shall be in accordance with General Specifications for Inspection of Material.

5. PREPARATION FOR DELIVERY

5.1 Packaging.-

5.1.1 Level A.- Sheets of plastic material shall be slip-sheeted-interleaved between each sheet with not less than 25-pound basis weight kraft paper in order to protect the sheets and prevent adherence to each other.

5.1.2 Level C.- Packaging shall be in accordance with the manufacturer's commercial practice. Each sheet shall be slip-sheeted to prevent adherence of sheets to each other.

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5.2 Packing.-

5.2.1 Level A.- The plastic sheets shall be loosely packed on end in fiber drums conforming to type II, grade A of Specification PPP-D-723.

5.2.2 Level B.- The plastic sheets shall be loosely packed on end in fiber drums conforming to type I, grade A of Specification PPP-D-723.

5.2.3 Level C.- The plastic sheets shall be packed in a manner which will insure acceptance and safe delivery at destination. Containers shall comply with the Consolidated Freight Classification Rules or other regulations as applicable to the mode of transportation.

5.3 Marking.- In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with Standard MIL-STD-129.

5.3.1 Precautionary marking.- Containers for sheets or rolls shall be prominently marked with the following:

KEEP COOL AND DRY

6. NOTES

6.1 Intended use.- The flexible, vinyl transparent sheet 0.060 inch in thickness is intended primarily for lenses for Navy gas masks. Sheet 0.040 inch in thickness is intended for windows in boat canopy curtains. The material may also be used wherever a tough, durable, flexible transparent plastic window is required.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Sheet size and thickness required (see 3.4.1).
(c) Whether domestic or overseas shipment is required; if domestic, the type required (see 5.1 and 5.2).

6.3 Fabricated parts or equipment.- Requirements applicable to fabricated parts or equipment incorporating this material should be specified in separate specifications or as part of contracts or orders.

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